RAJALAKSHMI ENGINEERING COLLEGE

V Y Aares 251801001-2025-AIDS V2

Started on	Wednesday, 23 September 2026, 10:13 PM
State	Finished
Completed on	Wednesday, 23 September 2026, 10:14 PM
Time taken	44 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include <stdio.h>
int count=0;
void Factor(int num)
 {
    for (int i = 1; i <= num;++i)
    {
     count++;
     if (num % i== 0)
        {
          count++;
          //printf("%d ", i);
        }
        count++;
     }
     count++;
  }
int main()
{
    int num;

    scanf("%d", &num);

    Factor(num);

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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